



11+ Maths: Fractions, Decimals and Percentages

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Write fractions in their simplest form unless the question says otherwise. Show working. You may not use a calculator.

Suggested time: 25 minutes · **Questions:** 20

1. Write $\frac{3}{12}$ in its simplest form. [1]
2. Convert $\frac{7}{10}$ to a decimal. [1]
3. Convert 0.45 to a percentage. [1]
4. Which is larger: $\frac{2}{5}$ or $\frac{3}{8}$? Show how you decide. [1]
5. Work out $\frac{2}{3} + \frac{1}{6}$. [1]
6. Work out $\frac{5}{8} - \frac{1}{4}$. [1]
7. Work out $\frac{3}{5} \times 10$. [1]
8. Work out $\frac{4}{9} \div \frac{2}{3}$. [1]
9. Write 1.2 as a mixed number and as an improper fraction. [1]
10. Find 15% of 80. [1]
11. A price increases from £40 to £50. What is the percentage increase? [1]
12. Decrease 240 by 10%. [1]
13. Write $\frac{18}{25}$ as a percentage. [1]
14. Put these in order, smallest first: 0.6, $\frac{2}{3}$, 55%, $\frac{5}{8}$. [1]
15. Find $\frac{3}{4}$ of £36. [1]
16. What fraction of 1 hour is 12 minutes? Give the answer in simplest form. [1]
17. A tank is 40% full. 15 litres are added and it is then 70% full. How many litres does the tank hold when full? [1]

18. Write $\frac{5}{8}$ as a decimal.

[1]

19. A test has 40 marks. Maya scores 32. What percentage did she get?

[1]

20. Which is equivalent to $\frac{6}{9}$? $\frac{2}{5}$ $\frac{4}{6}$ $\frac{3}{7}$ $\frac{5}{8}$

[1]

Answers and short explanations

Check your work only after you have finished. Equivalent wording is accepted where the meaning is the same.

1. **Answer:** $1/4$
Why: Divide numerator and denominator by 3: $3 \div 3 = 1$, $12 \div 3 = 4$.
2. **Answer:** 0.7
Why: 7 tenths is 0.7.
3. **Answer:** 45%
Why: Multiply by 100: $0.45 \times 100 = 45\%$.
4. **Answer:** $2/5$
Why: Common denominator 40: $2/5 = 16/40$, $3/8 = 15/40$. $16/40$ is larger.
5. **Answer:** $5/6$
Why: $2/3 = 4/6$. $4/6 + 1/6 = 5/6$.
6. **Answer:** $3/8$
Why: $1/4 = 2/8$. $5/8 - 2/8 = 3/8$.
7. **Answer:** 6
Why: $3/5 \times 10 = 30/5 = 6$.
8. **Answer:** $2/3$
Why: Keep, change, flip: $4/9 \times 3/2 = 12/18 = 2/3$.
9. **Answer:** $1 \frac{1}{5}$ and $6/5$
Why: $0.2 = 1/5$, so $1.2 = 1 \frac{1}{5} = 6/5$.
10. **Answer:** 12
Why: 10% of 80 = 8, 5% = 4, so 15% = 12.
11. **Answer:** 25%
Why: Increase = £10. $10/40 \times 100 = 25\%$.
12. **Answer:** 216
Why: 10% of 240 = 24. $240 - 24 = 216$.
13. **Answer:** 72%
Why: $18/25 = 72/100 = 72\%$.
14. **Answer:** 55%, $5/8$, 0.6, $2/3$
Why: As decimals: 0.55, 0.625, 0.6, 0.666... So 55%, $5/8$, 0.6, $2/3$.
15. **Answer:** £27
Why: $36 \div 4 = 9$, $9 \times 3 = 27$.
16. **Answer:** $1/5$
Why: $12/60 = 1/5$.
17. **Answer:** 50 litres
Why: The extra 30% is 15 litres, so 10% = 5 litres and 100% = 50 litres.
18. **Answer:** 0.625
Why: $5 \div 8 = 0.625$.
19. **Answer:** 80%
Why: $32/40 = 0.8 = 80\%$.

20. Answer: 4/6

Why: 6/9 and 4/6 both simplify to 2/3.

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